Lesson 6-4 · Quick

Logarithmic Functions + DOK-3 Sales Revenue Inverse (Item 28)

Klimsara-adapted · Student-centered · No Blooket

Do Now — Log translations + inverse setup

Algebra 2 · Lesson 6-4

DOK 1–2 | 7 min | *bridge to log inverses*

Three warm-up items (solo, 7 min)

1. **q31** — Translation multi-select for $h(x) = \ln(x+2) - 1$. Which statements are true?
2. **q5** — Graph $y = \log_4 x$. State domain, range, VA, x-intercept.
3. **q32 (SAT/ACT)** — Find the inverse of $f(x) = 5^{x+1}$. (Choices A–D.)

Launch — Example 3B: Inverse of a log function

DOK 2 | 8 min | *teacher models*

Rules Card — Log Key Features + Inverse Rules (keep on your page)

Key features of $y = \log_b x$: Domain $x > 0$; VA: $x = 0$; x-int: $(1, 0)$; passes through $(b, 1)$.

Translated form $y = \log_b(x - h) + k$: VA shifts to $x = h$; domain $x > h$.

Inverse rules:

- $y = a \cdot b^x \;\Rightarrow\; f^{-1}(x) = \log_b(x/a)$
- $y = \log_b(x - h) + k \;\Rightarrow\; f^{-1}(x) = b^{(x-k)} + h$

Composition check: $f(f^{-1}(x)) = x$

Launch — Example 3B walkthrough

DOK 2 | 8 min | *teacher models*

Ex 3B: Find the inverse of $g(x) = \log_7(x+5)$

1. Replace $g(x)$ with y: $y = \log_7(x+5)$
2. Swap x and y: $x = \log_7(y+5)$
3. Rewrite in exponential form: $7^x = y + 5$
4. Solve for y: $y = 7^x - 5 \quad \Rightarrow \quad g^{-1}(x) = 7^x - 5$
5. **Composition check:** $g(g^{-1}(x)) = \log_7(7^x - 5 + 5) = \log_7 7^x = x$ ✓

90-sec contrast — Ex 3A: inverse of $y = 3 \cdot 5^x$: same swap-and-solve, direction reversed $\to y = \log(x/3)$

Explore — Think-Pair-Share (25 min)

DOK 2–3 | 25 min | *student work*

5 items in order. Plan ∼10 min for Practice #28 (DOK-3). 45-min Wed F: skip q9.

Practice #28 · ★ DOK-3 SALES REVENUE INVERSE Sales revenue model. Find the inverse. Judge which equation form is easier for a given R.

Practice #21 · Inverse of $f(x) = 5^{x-3}$ Find the inverse. [Form B Q7 rehearsal]

Share / Summary

DOK 2 **5 min** *DOK-3 reveal + 6-5 bridge*

Sentence frame · Practice #28 "For Practice #28, I found the inverse by ___________.
For revenue $R =$ ____ dollars, the original form is easier because __________.
The inverse form is easier when ______________."

Bridge to Lesson 6-5 Log properties (product, quotient, power rules) let you expand or condense log expressions — making inverses and equations faster.

Exit Ticket — Inverse + Asymptote Algebra 2 · Lesson 6-4

DOK 1–2 | 5 min | *not graded*

Complete on your exit ticket

1. Find the inverse of $f(x) = 3 \cdot \log_5(x-2) + 1$.
2. State the vertical asymptote of f.
3. State the horizontal asymptote of f^{-1}.
4. One biggest thing I learned today: ____________________.